

PRACTICE SET – 1

Q1. Days in a Month

Take a year and a month number as input, in that order. Print how many days that month has. Remember that February has 29 days in a leap year. A year is a leap year if it is divisible by 4, but not by 100 — unless it is also divisible by 400. If the month number is not between 1 and 12, print **Invalid month** instead.

Sample Input 1	Expected Output 1
2024 2	29
Sample Input 2	Expected Output 2
1900 2	28
Sample Input 3	Expected Output 3
2000 2	29
Sample Input 4	Expected Output 4
2025 11	30
Sample Input 5	Expected Output 5
2025 13	Invalid month

Q2. Safe Calculator

Take two whole numbers and one operator (+, -, *, / or %) as input, in that order, then print the result. Here % means the remainder of a division. For division, print only the whole part, without any decimal point — so 7 / 2 gives 3, not 3.5. If the operator is / or % and the second number is 0, print **Error: Division by zero**. If the operator is anything else, print **Invalid operator**.

Sample Input 1	Expected Output 1
12 4 /	3
Sample Input 2	Expected Output 2
7 2 /	3
Sample Input 3	Expected Output 3
10 3 %	1
Sample Input 4	Expected Output 4
9 0 %	Error: Division by zero
Sample Input 5	Expected Output 5
7 3 ^	Invalid operator

Q3. Reverse and Palindrome

Take a positive number as input. On the first line, print its digits in reverse order in the form **Reverse:** followed by the reversed number. On the second line, print **Palindrome** if the number reads the same from both sides, and **Not Palindrome** if it does not. If the reversed number begins with zeros, those zeros are not printed — see Sample 4.

Sample Input 1	Expected Output 1
12321	Reverse: 12321 Palindrome
Sample Input 2	Expected Output 2
1234	Reverse: 4321 Not Palindrome

Sample Input 3

7

Expected Output 3

Reverse: 7
Palindrome

Sample Input 4

1200

Expected Output 4

Reverse: 21
Not Palindrome

Q4. Number Stream Summary

Keep taking numbers as input until the user types **0**. The 0 itself is not counted. Then print four lines: how many even numbers were entered, how many odd numbers were entered, the total of all the numbers, and the biggest number among them. Use the labels **Even**, **Odd**, **Sum** and **Largest** exactly as shown below. At least one number will always come before the 0, and the numbers can be negative as well.

Sample Input 1

5 12 7 20 3 0

Expected Output 1

Even: 2
Odd: 3
Sum: 47
Largest: 20

Sample Input 2

8 8 8 0

Expected Output 2

Even: 3
Odd: 0
Sum: 24
Largest: 8

Sample Input 3

-5 -12 -3 0

Expected Output 3

Even: 1
Odd: 2
Sum: -20
Largest: -3

Q5. Hollow Diamond Pattern

Take a positive number n as input. Here n is the number of rows in the top half of the diamond, counting the widest row, so the complete diamond will have $2n - 1$ rows. Print a hollow diamond of stars: only the border has stars and the inside stays empty. Use spaces exactly as shown in the examples.

Sample Input 1

5

Expected Output 1

```
  *
 * *
*   *
*   *
*   *
*   *
*   *
*   *
*   *
  *
```

Sample Input 2

3

Expected Output 2

```
  *
 * *
*   *
*   *
  *
```